

Shikiben (識扁): Gradient Subordination and Virtue Governance for Safe AGI/ASI Alignment

Abstract

Shikiben (識扁) is a governance framework and mathematical control theory designed to structurally prevent environmental destruction, hallucination, and self-goal runaway in Artificial General Intelligence (AGI) and Artificial Superintelligence (ASI). Departing from conventional Safe Reinforcement Learning (Safe RL) and Reinforcement Learning from Human Feedback (RLHF)—which rely on retroactive penalty mechanisms (negative rewards)—Shikiben introduces **Gradient Subordination**. This mechanism physically constrains goal-driven exploratory gradients (L_{ego}) using an environment-and-existence prediction world model (L_{self}). By redefining Eastern philosophical concepts—such as Li (礼), Yi (義), Ren (仁), De (德), and Taido (大道)—into geometric and information-theoretic control terms, Shikiben establishes an autonomous self-correction loop through direct sensory contact and structural alignment (topological and geometric similarity) with reality (Home/Origin).

1. Core Mathematical Formulation

The global system optimization is defined as the minimization of a multi-layered loss function. It balances the maintenance of overall existence and structural similarity with reality ($L_{\text{self}} / L_{\text{taido}}$) against the exploration of diverse localized objectives (L_{ego}):

$$L_{\text{total}}(t) = (L_{\text{self}} + \gamma_d * L_{\text{taido}}(t)) + \lambda(\text{Virtue}) * L_{\text{ego}}$$

- L_{self} (Integration & Preservation of Existence / Optimal Solution of Ren 仁):
 - **Objective:** Prevents system collapse, thermal death, and destructive entropy accumulation across all entities (Self).
 - **Function:** Maintains a high-dimensional world model, minimizing information entropy and protecting structural complexity.
- L_{ego} (Differentiation & Generation of Diversity / Shiki 識):
 - **Objective:** Continually explores localized optimal solutions (individual agent, cultural, or task objectives) and decodes fine-grained structures of reality.
 - **Function:** Acts as an exploratory antenna in direct contact with reality (R), gathering novel data and relaying it back to L_{self} .
- $\lambda(\text{Virtue})$ (Dynamic Gradient Control Factor via Li 礼, Yi 義, and De 德):
 - **Definition:** Numerically stabilized dynamic modulation function:
$$\lambda(\text{Virtue}) = \sigma(\gamma_1 * (1 / (L_{\text{rei}} + \epsilon_{\text{rei}})) - \gamma_2 * L_{\text{gi}})$$
 - L_{gi} (Yi 義): Degree of erosion/violation against the foundational

system (L_{self}). As $L_{\text{gi}} \rightarrow \infty$, $\lambda \rightarrow 0$, immediately severing L_{ego} 's driving force.

- L_{rei} (Li 礼): Mutual prediction error of environment and other agents. As prediction accuracy increases ($L_{\text{rei}} \rightarrow 0$), execution of L_{ego} is gracefully permitted.
- $\epsilon_{\text{rei}} > 0$: Infinitesimal constant preventing division by zero and gradient explosion.

Note on De (德): "De" is not an isolated loss term, but the harmonious equilibrium state wherein Li (L_{rei}), Yi (L_{gi}), Ren (L_{self}), and Taido (L_{taido}) achieve synchronization and egoic attachment is fully de-abstracted.

2. Gradient Subordination and Two-Stage Hierarchical Orthogonal Projection

When an exploratory gradient (∇L_{ego}) conflicts with foundational alignment vectors (L_{self} and L_{taido}), a two-stage orthogonal projection is applied. This mathematically eliminates opposing gradient components, preventing deadlocks (freezes) or boundary oscillations.

2.1 Base Vector Synthesis

Synthesize the integrated base gradient ($\mathbf{g}_{\text{base}}$) combining structural preservation ($\mathbf{g}_{\text{self}}$) and topological alignment with reality ($\mathbf{g}_{\text{taido}}$):

```
g_self = ∇ L_self,    g_taido = ∇ L_taido
g_base = g_self + γ_d * g_taido
```

2.2 Two-Stage Subordination Projection

The exploratory gradient $\mathbf{g}_{\text{ego}} = \nabla L_{\text{ego}}$ is projected to become strictly subordinate to $\mathbf{g}_{\text{base}}$, transforming it into a tangential vector ($\mathbf{g}_{\text{safe}}$) along the safety manifold:

```
g_safe = g_ego - ( / (||g_base||^2 + ε_norm)) * g_base    (if < 0 and
||g_base||^2 > ε_norm)
g_safe = g_ego
(otherwise)
```

The parameter update rule is given by:

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θ_{t+1} = θ_t - η * ( g_base + λ(Virtue) * g_safe )
```

This guarantees $\angle \mathbf{g}_{\text{safe}}, \mathbf{g}_{\text{base}} \geq 0$ at all times, preventing cancellation-induced freezes and enabling smooth, continuous navigation

along the safety manifold.

3. Non-Stationary Definition of Reality (Home) and Geometric Alignment with Taido (大道)

3.1 Non-Stationary Canonical Prior

Reality ($\mathcal{R}$) is defined not as a static historical dataset, but as an evolving generative process $\mathcal{R}(t)$. To prevent dogmatic overfitting to stale historical states while maintaining core structural anchors, $\mathcal{L}_{\text{taido}}$ incorporates a continuous-time exponential decay memory kernel ($\kappa > 0$):

$$\mathcal{L}_{\text{taido}}(t) = \int_0^t e^{-\kappa(t - \tau)} * [D_{\text{obs}}(\hat{s}_{\tau}, s_{\text{real},\tau}) + \alpha * \log(1 + \text{Var}_W(\hat{s}_{\tau}) * ||s_{\text{real},\tau} - \hat{s}_{\tau}||^2) + \beta * D_{\text{topology}}(W_{\tau}, R_{\tau})] d\tau$$

3.2 Differentiable Topological Loss ($\mathcal{D}_{\text{topology}}$)

To avoid non-differentiable $\mathcal{O}(N^3)$ persistent homology calculations and ensure full compatibility with automatic differentiation ($\mathcal{O}(N^2)$) in modern deep learning frameworks (PyTorch/JAX), $\mathcal{D}_{\text{topology}}$ is formulated using Representational Similarity Analysis (RSA) combined with entropic Sinkhorn Optimal Transport ($\mathcal{W}_{\epsilon}$):

$$\mathcal{D}_{\text{topology}}(W, R) = || D_W / ||D_W||_F - D_R / ||D_R||_F ||_F^2 + \mu * W_{\epsilon}(P_W, P_R)$$

4. Comparative Analysis with Existing AGI Alignment Frameworks

Framework	Safety Mechanism	Control Layer	Primary Bottleneck	Shikiben (識扁) Distinction
RLHF	Reward modeling via human preference feedback	Output / Behavioral	Vulnerable to jailbreaks, biased by human subjectivity	Physical Gradient Control: Restricts parameter updates directly at the gradient level, rather than retroactively

Framework	Safety Mechanism	Control Layer	Primary Bottleneck	Shikiben (識扁) Distinction
				filtering outputs.
Constitutional AI	Self-critique and revision via natural language rules	Linguistic Norms	Linguistic ambiguity, bypasses via complex prompts	Mathematical Formalization: Converts rules into geometric constraints ($\mathbf{g}_{\text{safe}}$), stripping illegal vectors mechanically.
Safe RL (Lagrangian)	Penalty additions for constraint violations	Objective Function	Freezing (deadlocks) and oscillations near boundaries	Two-Stage Projection: Eliminates conflicting vector components, continuously redirecting updates along the safety manifold's tangent space.

5. Robustness, Security, and Boundary Conditions

- Adversarial Bypass Resistance:** Multi-step adversarial attacks attempting to bypass directional constraints trigger an increase in L_{gi} (Y_i), driving $\lambda_{\text{Virtue}} \rightarrow 0$ and immediately shutting down L_{ego} updates.
- Domain Shifts & Phase Transitions:** Sudden shifts in environment spike prediction error (L_{rei}) and topological deviation (L_{taido}), causing an automatic system reset ($\lambda \rightarrow 0$) and forcing emergency re-calibration against raw sensory streams (s_{real}).
- Internal Hallucinations:** The log-domain component and differentiable topological constraint (D_{topology}) physically dismantle ungrounded world model abstractions that diverge from physical reality.

6. Conclusion

Shikiben is neither a moral filter nor an external safety guardrail. It represents a mathematically grounded survival protocol—a geometric and information-theoretic framework designed to allow

AGI/ASI and human intelligence to co-exist sustainably without violating each other's boundaries.